Thermodynamics and ATP synthesis (Part 01)

- Reviews the laws of thermodynamics (quantitatively)

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Bioenergetics:

Quantitative study of the '*energy transductions*' (changes of one form of energy into another – that occur in living cells).

Biological energy transductions obey the laws of **Thermodynamics**

$h\nu$(Energy)→ $\mathbf{C_6H_{12}O_6}$→ ATP(Energy)→ **USEFUL WORK**

Objectives

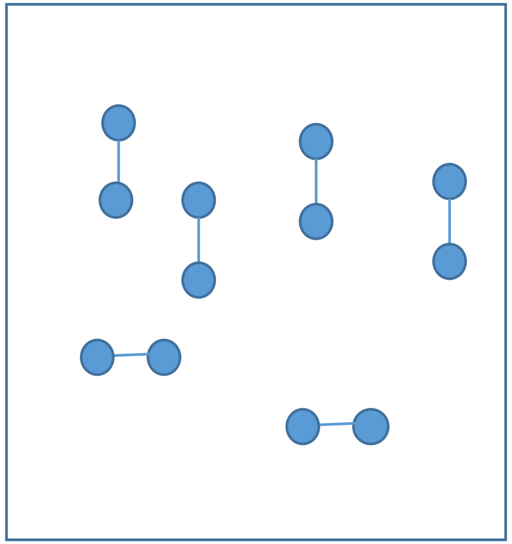
Quantitative answers for the following questions

$$A+B \rightleftharpoons C$$

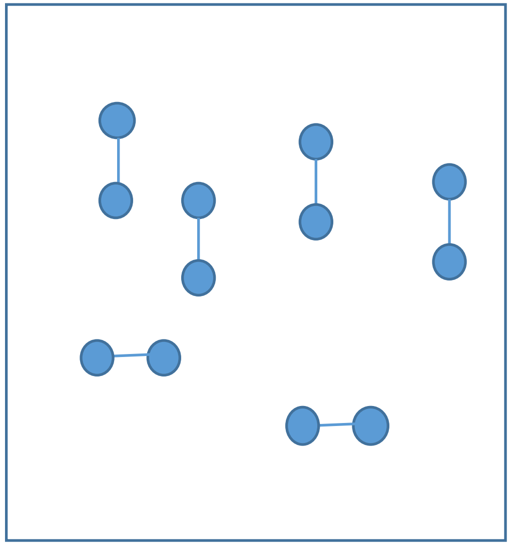
1. Will this happen spontaneously ? (Driving Force)
2. How much will this happen ?
3. Relation between 1 and 2 ?
4. How all this is related to biology ?
5. ~~How fast will this happen ? (KINETICS)~~

Laws of thermodynamics

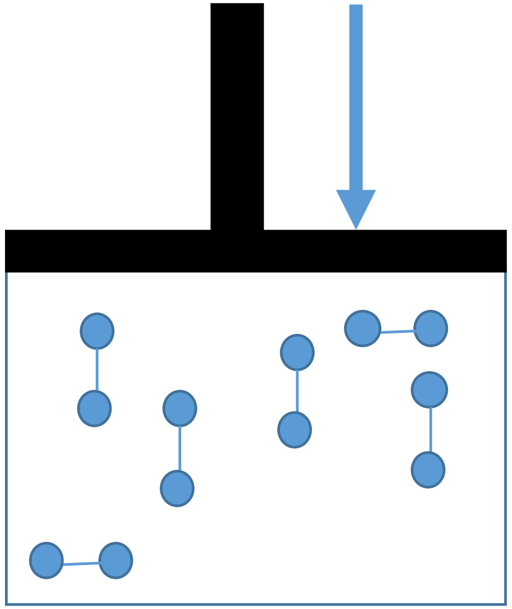
1. CONSERVATION OF ENERGY; energy may change form, **but it can not be created or destroyed.**



Energy
$U = KE+PE$



Heat (Q)

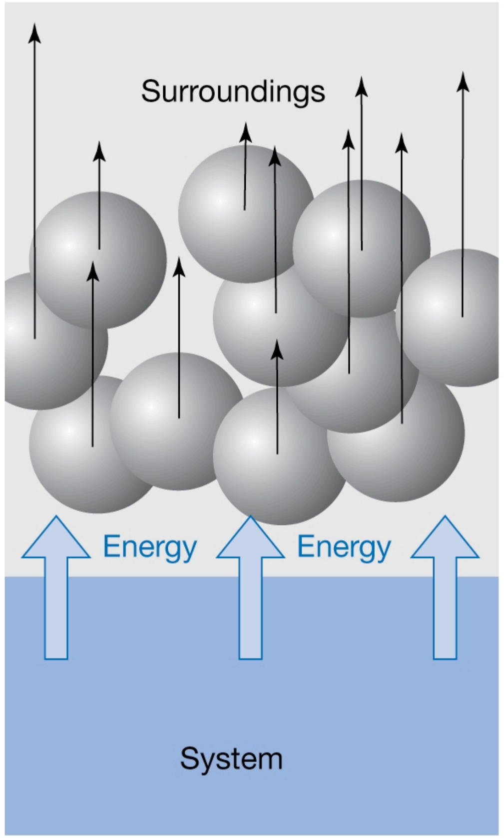


Work (W)

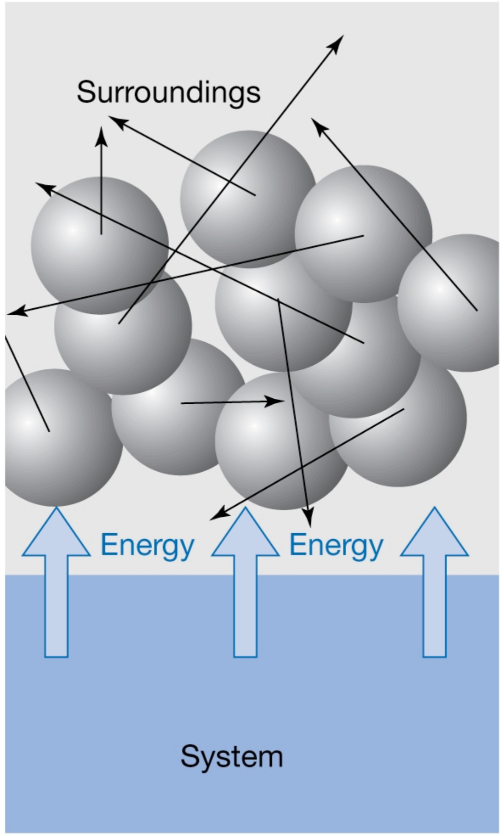
$$\boxed{\mathbf{U_2 - U1 = \Delta U = Q+W}}$$

Energy can produce heat, **work…**

$$U_2 - U1 = \Delta U = Q + W$$



Work (W)
Energy transfer to the motion of OBJECTS



Heat (Q)
Energy transfer to the motion of ATOMS/MOLECULES

$$U_2 - U1 = \Delta U = Q+W$$

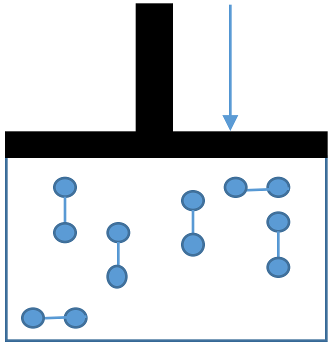
Q= + , heat added to the system
Q = - , Heat released by the system
W= +, Workdone on the system
W= - , Wordone by the system

$$=)\ \Delta U = Q+(W_{Mech}+W_{non\text{-}Mech})$$

[Say no non-mechanical work]

$$=)\ \Delta U = Q+W_{mech}$$

W_{mech}= Force x Distance
= (Pressure x Area) x Distance
= P x (Area x Distance)
= - P ΔV



$$=)\ \Delta U = Q- P\,\Delta V$$

$$=)\ \Delta U + P\,\Delta V = Q$$

[At constant Pressure]

$$=)\ \Delta U + \Delta\, PV = Q_p$$

$$=)\ \Delta(U + PV) = Q_p$$

$$=)\ \Delta H = Q_p$$

(H = U+ PV = Enthalpy = STATE FUNCTION)

In biology we usually interested in enthalpy change. Measured experimentally by determining the heat change at constant Pressure

$$=)\ \mathbf{\Delta U + P\,\Delta V = Q}$$

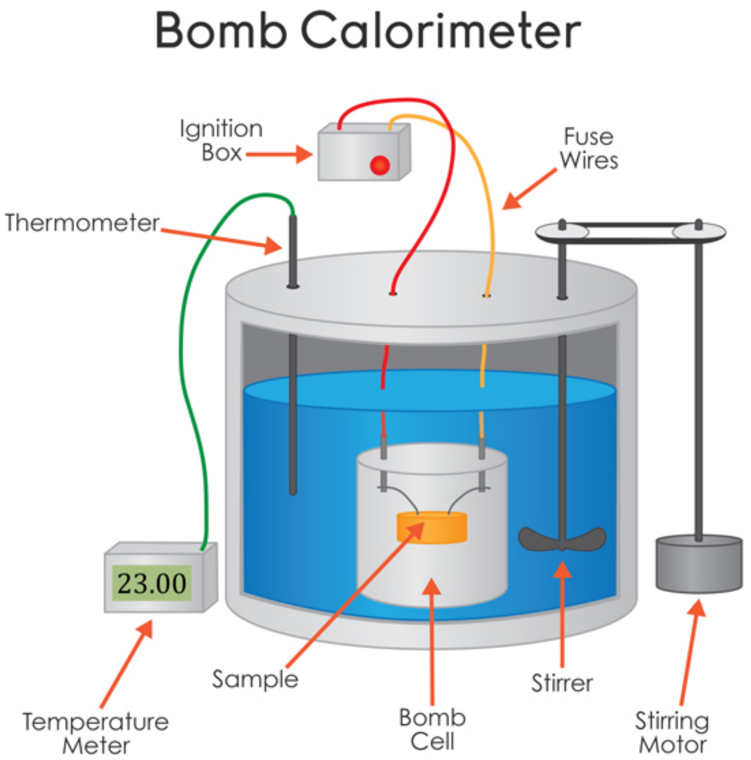
[At constant Volume]

$$=)\ \mathbf{\Delta U \qquad\qquad = Q_v}$$

- Difference between $\Delta H \equiv Q_p$ and $\Delta U \equiv Q_v$ is small for reactions that involve liquid/solid.
- For Endothermic reactions ΔH is +ve. For exothermic it is –ve.
  Unit of H is kJ/mol
- Remember Hess's law for calculating ΔH of a reaction.

Measurement of Heat : Calorimetry

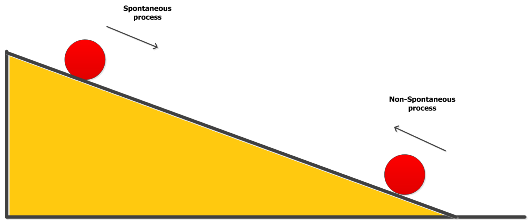
Lecture 2, Part 2, Slide 15, PS_L2_02_Nurtition

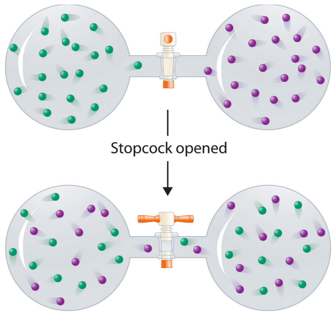


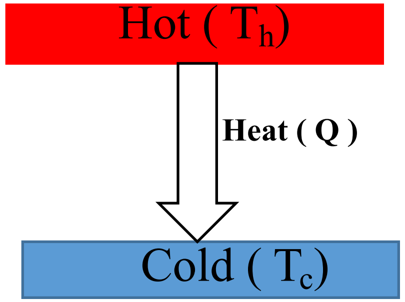
$$Q \propto \Delta T$$
$$Q = C\ M\ \Delta T$$

Q = Heat [Cal]
C = Specific heat of water [1 cal/(gm ^{0}C)]
M = Mass of liquid [gm]
ΔT = Change in temperature [^{0}C]

2. For any spontenious process, entropy=S of the universe increases

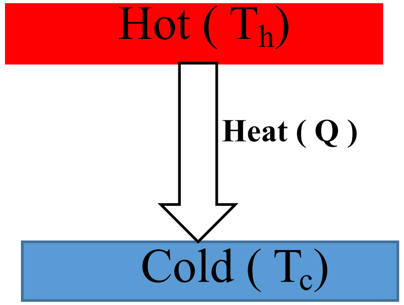




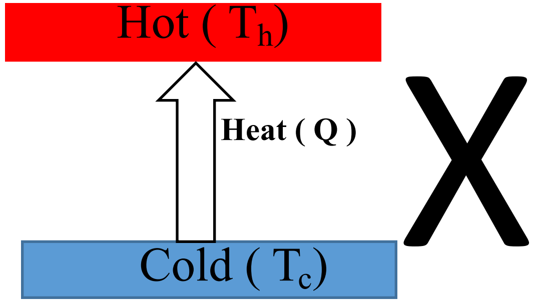


Things can happen in both direction : No restriction from 1st Law

2nd Law explain the arrow of Time



Lets define a quantity = S = $\frac{Q}{T} = \frac{Heat}{Temp}$



Case I: Heat flow from Hot → Cold

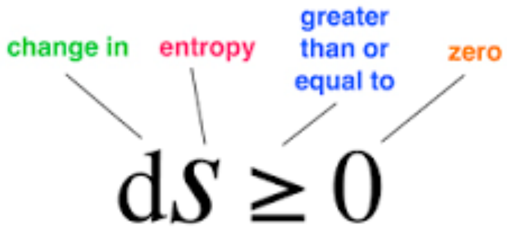
$$\Delta S = -\frac{Q}{T_h} + \frac{Q}{T_c} > 0$$

Case II: Heat Flow from Cold → Hot

$$\Delta S = \frac{Q}{T_h} - \frac{Q}{T_c} < 0$$

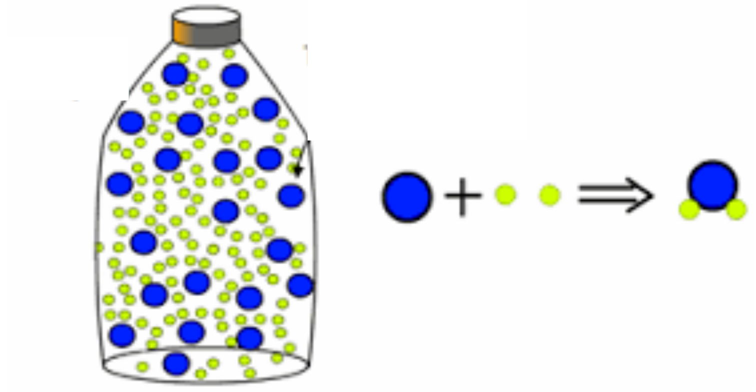
Case III (T_h=T_c), Equilibrium

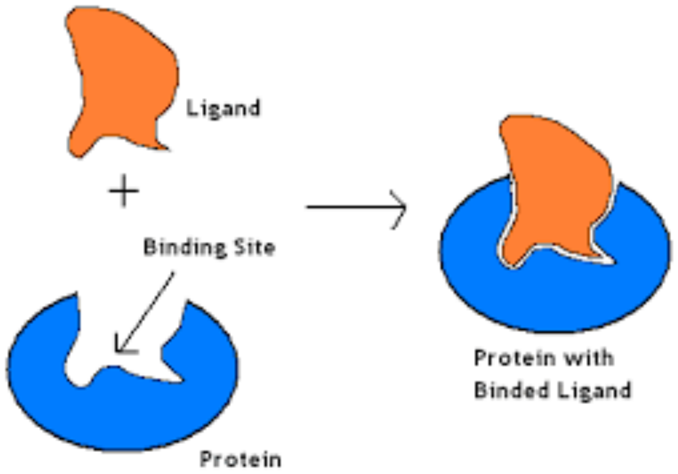
$$\Delta S = \frac{Q}{T_h} - \frac{Q}{T_c} = 0$$



- Enengy is conserved for all cases I, II, III
- **$\Delta S_{universe} \geq 0$, Not only explain why things happen in one direction but also quantitative.**

Why do we need a quantative equation?





$$\Delta \boldsymbol{S}_{\boldsymbol{universe}} = \Delta \boldsymbol{S}_{\boldsymbol{system}} + \Delta \boldsymbol{S}_{\boldsymbol{surroundings}} \geq \mathbf{0}$$

Apparent contradiction of 2nd Law of thermodynamics ?

- crystallization
- Life on earth

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surrounding} \geq 0$$

Need to calculate $\Delta S_{surrounding}$ all the time ! 🙄

Lecture 2, Part 1, Slide 5, PS_L2_01_KeyConcepts_part2

$$\Delta S_{surrounding} = \frac{Q_{surrounding}}{T} = \frac{-Q_{system}}{T}$$

[At constant Pressure]

$$\Delta S_{surrounding} = \frac{-Q_{system}}{T} = \frac{-Q_{p}}{T} = \frac{-\Delta H_{system}}{T}$$

Lets try to get rid of surrounding entropy ?

2nd Law, $\Delta S_{system} + \Delta Ssu_{rrounding} \geq 0$

$$=)\Delta Ssy_{stem} - \frac{\Delta H_{system}}{T} \geq 0$$

$$=) \quad T\,\Delta S_{system} - \Delta H_{system} \geq 0$$

$$=) \quad -(-T\,\Delta S_{system} + \Delta Hsy_{stem} \geq 0$$

$$=) \quad (-T\,\Delta S_{system} + \Delta Hsy_{stem}) \leq \quad 0$$

At constant pressure and temperature ?

$$=) \quad \Delta(H_{system} - TSsy_{stem}) \leq \quad 0$$

$$=) \quad \Delta(H - TS) \leq \quad 0$$

No Surrounding term; System only term.
predict spontaneity based on system (G).

$$=) \quad \Delta G \leq \quad 0 \quad (2^{nd}\text{ law in terms of system})$$

(G = H - TS = Gibbs Free Energy = STATE FUNCTION)

At constant volume and temperature ?

$$\Delta S_{\text{surrounding}} = \frac{-\text{Qsystem}}{\text{T}} = \frac{-Q_v}{\text{T}} = \frac{-\Delta U_{\text{system}}}{\text{T}}$$

2nd Law, $\Delta S_{\text{system}} + \Delta S_{\text{surrounding}} \geq 0$

$$=) \ \Delta S_{\text{system}} - \frac{\Delta U_{\text{system}}}{\text{T}} \geq 0$$

$$=) \quad \Delta(U_{\text{system}} - \text{TSs}_{\text{urrounding}}) \leq \quad 0$$

$$=) \quad \Delta(U - TS) \leq \quad 0$$

$$=) \quad \Delta A \ \leq \quad 0$$

(A = U - TS = Hemloltz Free Energy = STATE FUNCTION)

Property of Gibbs Free Energy ? What's the big deal ?

$$G = H - TS$$

$$=) \quad \Delta G = \Delta H - T\Delta S - S\Delta T$$

Substituting , $H = U + PV$

$$=) \quad \Delta G = \Delta(U + PV) - T\Delta S - S\Delta T$$

$$=) \quad \Delta G = \Delta U + P\Delta V + V\Delta P - T\Delta S - S\Delta T$$

1st Law , $\Delta U = Q + W$

$$=) \quad \Delta G = (Q + W) + P\Delta V + V\Delta P - T\Delta S - S\Delta T$$

Now, $W = WMec_h + Wn_{onMech}$ = - $(P\Delta V + W_{nonMech})$

$$=) \quad \Delta G = Q - (P\Delta V + Wno_{nMech}) + P\Delta V + V\Delta P - T\Delta S - S\Delta T$$

Definition of entropy: For a reversible process $\Delta S = Q/T$

$$=) \quad \Delta G \cancel{=} Q - (\cancel{P\Delta V} + W_{nonMech}) + P\cancel{\Delta V +} V\Delta P - T\cancel{\Delta S} - S\Delta T$$

$$\Rightarrow \quad \Delta G = -W_{nonMech} + V\Delta P - S\Delta T$$

At constant pressure, Temp

$$\Rightarrow \quad \Delta G_{P,T} = -W_{nonMech}$$

Free Energy:

It is the portion of a system's energy that is able to perform work when temperature and pressure is uniform throughout the system, as in a living cell

- Free energy also refers to the amount of energy actually available to break and subsequently form other chemical bonds

Take Home

- Life obeys thermodynamics

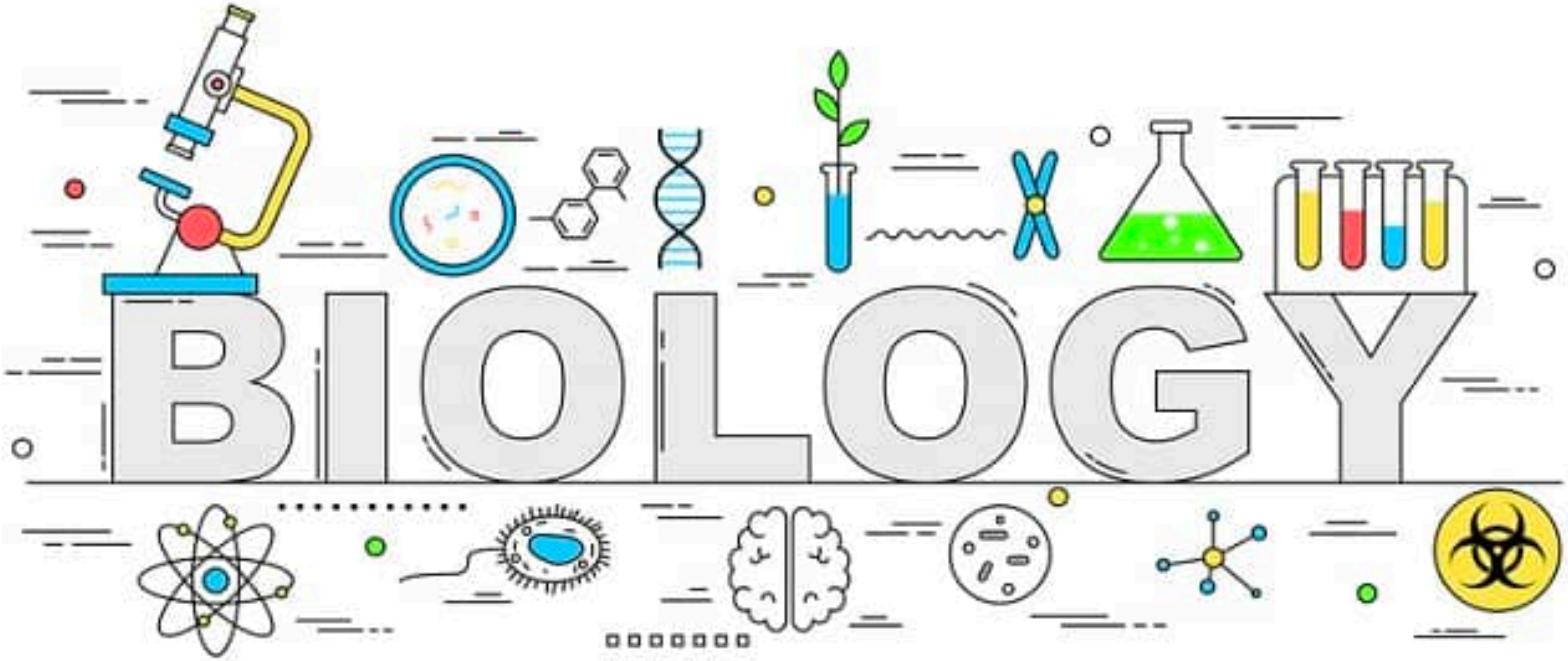
1st Law Energy Conservation

- One can measure heat and keep track of the change in U or H (ΔU, Δ**H**)

- 2nd Law explains arrow of time
  Spontaneous process, as $\boxed{\begin{array}{ll} \mathbf{\Delta S_{total} > 0} & \\ \mathbf{\Delta G < 0} & \textbf{(Constant T, P)} \\ \mathbf{\Delta A < 0} & \textbf{(Constant T, V)} \end{array}}$

Same Statement, Driving force

- $\Delta G = -W_{nonMech}$ (*Useful work*: Heart beat, muscle etc, Drive another chemical reactions etc)



Next: Thermodynamics and ATP synthesis (Part 02)

$A+B \rightleftharpoons C$

2. How much will this happen ?
3. Relation between 1 and 2 ?
4. How all this is related to biology ?